

ICAI Level 2 Capstone Project Report

SmartAudit Pro - Audit Sample Selector

Project Type: Python Desktop Application

Domain: Statutory Audit / Audit Analytics

Prepared for: ICAI Level 2 Capstone Project

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1. Executive Summary

SmartAudit Pro is a local desktop application designed to demonstrate technology-assisted audit population analysis and sample selection. It accepts Excel or CSV transaction data, provides population analytics, supports four selection approaches, displays dashboard charts, provides an indicative sample-size calculator, and exports Excel and PDF reports.

The project addresses a practical audit problem: converting a raw transaction population into a documented, reproducible selection workflow while preserving the auditor's responsibility for professional judgement.

2. Problem Statement

Audit teams frequently receive large transaction dumps. Manual spreadsheet-based analysis may require repeated sorting, filtering, randomisation, copying and documentation. This can make the selection process time-consuming and difficult to reproduce.

The project therefore seeks to create a simple desktop application that consolidates population analysis, sample selection, rule-based risk identification and reporting.

3. Objectives

The objectives are to: (a) import Excel/CSV populations; (b) provide basic population analytics; (c) demonstrate random, systematic, stratified and risk-based selection; (d) visualise population characteristics; (e) provide an educational sample-size calculator; and (f) generate documented Excel/PDF outputs.

4. Scope

The prototype focuses on transaction-level populations such as journal entries, expense ledgers or purchase registers. It assumes that the user has already obtained an appropriate population and remains responsible for validating its completeness and relevance.

It is not an audit-management system and does not independently determine whether sufficient appropriate audit evidence has been obtained.

5. Functional Modules

Module	Function
Data Import	Loads .xlsx/.xlsm or .csv transaction populations.
Field Mapping	Maps Amount, Date, User and Narration fields.
Population Analytics	Displays population count, absolute value, average, maximum amount and risk-rule count.
Random Sampling	Selects N items using a reproducible pseudo-

	random seed.
Systematic Sampling	Uses a random start and approximately equal intervals.
Stratified Sampling	Divides the value-sorted population into three strata and allocates samples.
Risk-based Selection	Ranks transactions using illustrative risk rules and transaction value.
Dashboard	Shows value distribution and rule-based risk distribution.
Sample-size Calculator	Provides an indicative finite-population attribute sample size.
Reporting	Exports Excel and PDF audit reports.

6. Sampling Logic

Random selection uses Python's pseudo-random generator with a user-defined seed. A fixed seed allows the same population and parameters to reproduce the same selection.

Systematic selection calculates an approximate interval as population size divided by required sample size and applies a random starting point.

Stratified selection sorts transactions by absolute value, divides the population into three value bands, and allocates the requested sample broadly in proportion to stratum size.

Risk-based selection is targeted testing rather than statistical sampling. Transactions are ranked using rule-based points and absolute value.

7. Illustrative Risk Rules

Rule	Illustrative score
Amount at/above threshold	4
Round amount near multiples of ₹1,00,000	2
Weekend posting	2
Posting from 25-31 March	2
Sensitive narration keywords	2

The above rules are illustrative only. In an actual audit, risk criteria should be tailored to the entity, process, fraud risks, controls and audit strategy.

8. Sample-size Calculator

The calculator uses a finite-population proportion formula with population size, confidence level, tolerable deviation and expected deviation. It is an educational planning aid and is not represented as an ICAI or SA 530 prescribed sampling table.

9. Technology Architecture

Technology	Use
Python	Application logic
Tkinter	Desktop user interface
openpyxl	Excel import/export
matplotlib	Dashboard charts
ReportLab	PDF report generation

10. Demonstration Dataset

The repository includes SmartAudit_Demo_Population.xlsx containing 500 dummy GL transactions. It deliberately contains high-value, round-value, year-end and sensitive-narration transactions so that each feature can be demonstrated without using confidential client information.

11. Audit Standards Context

The project is conceptually relevant to SA 530, Audit Sampling, because it demonstrates different methods of selecting items from an audit population. The auditor must nevertheless determine whether sampling is appropriate, define the population and sampling unit, determine sample size, perform procedures on selected items, investigate deviations or misstatements and evaluate the results.

The rule-based risk module can support the auditor's analysis of unusual transactions, but it does not replace the risk assessment required by applicable Standards on Auditing.

12. Key Controls and Limitations

- Population completeness is not independently verified by the application.
- Risk rules are illustrative and not entity-specific.
- The sample-size calculator is indicative and not an SA 530 prescribed table.
- Risk-based selection is targeted testing, not statistical sampling.
- The prototype does not evaluate audit exceptions or conclude on the population.
- Users should reconcile imported data and validate outputs before any real engagement use.

13. Future Enhancements

- Monetary Unit Sampling and user-defined stratification.
- Configurable risk-rule builder and duplicate detection.
- Population reconciliation and data-quality checks.
- Saved engagement configurations and audit-trail logs.
- Role-based access, encryption and secure project storage.

14. Conclusion

SmartAudit Pro demonstrates that a focused desktop application can combine data import, audit-oriented analytics, multiple selection methods, visualisation and documentation in a simple workflow. Its principal value as a capstone project is the integration of accounting/audit domain knowledge with practical Python development while clearly preserving professional judgement.